



The Impact of Body Composition on Climbing Performance: Correlations Between Muscle Mass, Grip Strength, and Psychological Factors in Elite and Recreational Female Rock Climbers



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Abstract

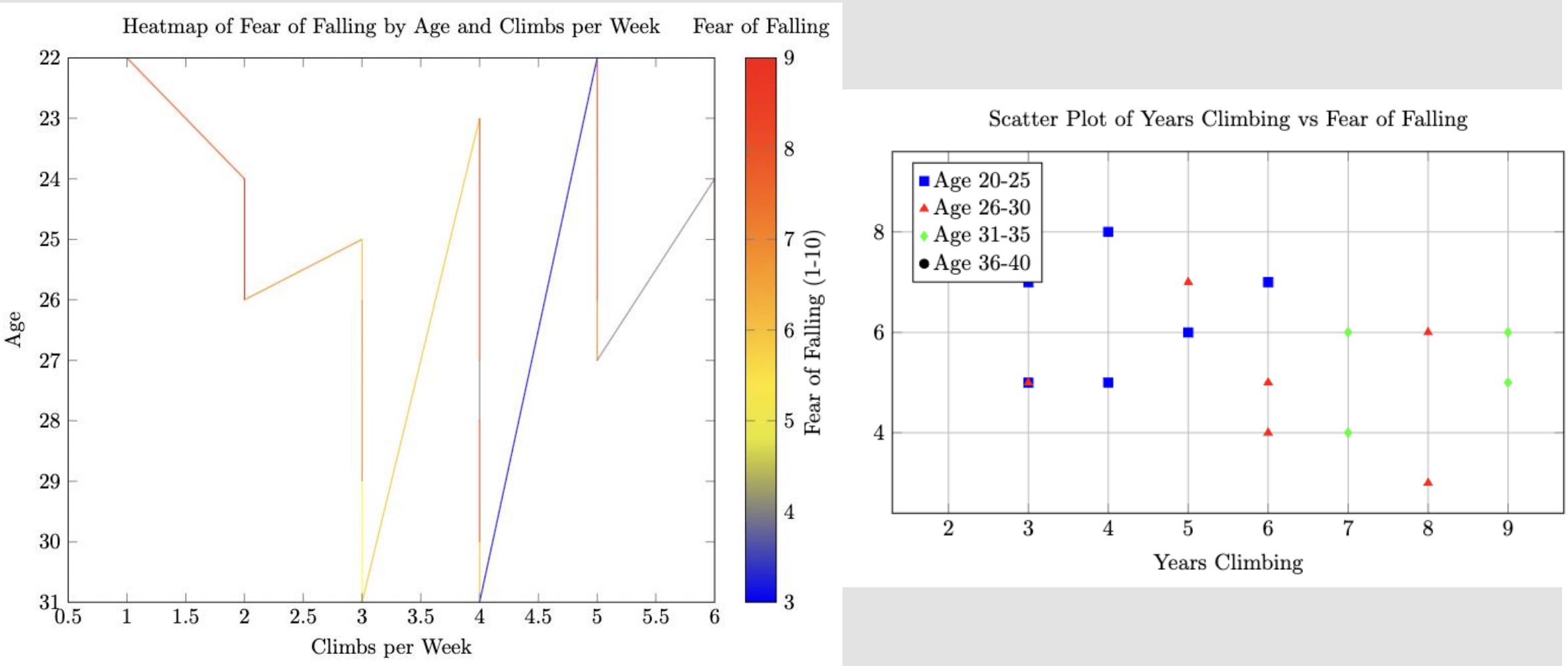
Despite climbing's popularity and an increasing number of female participants, there are limited anthropometric and performance data for this demographic. Previous research highlights the importance of body composition, particularly the ratio of muscle mass to body fat, in climbing performance. This study synthesizes findings from existing DEXA-based studies and supplements them with survey data and performance testing to evaluate these variables. Notably, the research reveals that higher muscle mass correlates positively with grip strength, which, in turn, is a critical factor in climbing performance. Psychological factors, such as fear of falling, also significantly affect climbers' performance, with higher fear levels observed in those who climb more frequently. The findings emphasize the need for more focused research on female climbers and highlight the limitations of current studies, such as the lack of control conditions and gender-specific data. This research underscores the importance of a holistic approach to climbing training, incorporating both physical and psychological aspects, and calls for future studies to refine and expand upon these findings.



Me with @BrownGirlsClimb!



Qualitative Survey Visualizations:



Methodology

This study employs a mixed-methods approach, integrating both qualitative and quantitative research techniques to offer a comprehensive understanding of the factors influencing female rock climbers' performance.

The quantitative component draws on data from 5 similar experiments of 55 experienced female climbers, categorized into lower advanced (ADV-L), higher advanced (ADV-H), and elite (ELT) groups based on self-reported abilities. This dataset includes detailed assessments of climbing experience, body composition through DEXA scans, flexibility, and strength measurements. Grip strength, a critical factor in climbing ability, is measured using a dynamometer to evaluate its impact on climbing proficiency.

Qualitative insights are derived from a structured survey distributed to members of climbing organizations, including BrownGirlsClimb, mainly based in New York and California. The survey (with a yield of 26 responses) captures psychological factors such as fear of falling, self-efficacy, risk-taking behaviors, and emotion regulation strategies. By combining these approaches, the study aims to provide valuable insights into the physiological and psychological factors that contribute to climbing success among recreational and elite female climbers.

Participant	Years Climbing	Climbs/Week	Climbing Types	Age	Fear of Falling (1-10)	Anxiety Management	Strength Training
1	2	3	Bouldering, Sport	25	6	Visualization	Fingerboard, Campus Board
2	3	4	Sport, Trad	30	7	Breathing Exercises	Weight Lifting, Core
3	7	2	Bouldering	27	4	Mental Rehearsal	Fingerboard
4	4	5	Sport, Bouldering	22	8	Relaxation Techniques	Campus Board, Hangboard
5	6	3	Sport	29	5	Positive Self-Talk	Weight Lifting, Yoga
6	2	6	Bouldering, Trad	24	9	Meditation	Fingerboard, Climbing Specific
7	8	4	Sport, Bouldering	31	3	Visualization	Core, Fingerboard
8	5	2	Trad	28	6	Breathing Techniques	Hangboard, Yoga
9	3	3	Bouldering	26	7	Relaxation	Weight Lifting, Core
10	9	1	Sport, Bouldering	32	5	Mental Rehearsal	Fingerboard, Campus Board
11	4	4	Sport	23	8	Positive Self-Talk	Weight Lifting
12	7	3	Bouldering, Trad	30	6	Visualization	Fingerboard, Hangboard
13	6	5	Sport, Bouldering	27	4	Meditation	Campus Board, Core
14	5	2	Trad	26	7	Breathing Exercises	Weight Lifting, Core
15	3	6	Bouldering, Sport	25	5	Relaxation Techniques	Weight Lifting, Core
16	8	3	Sport, Bouldering	29	6	Positive Self-Talk	Hangboard, Fingerboard
17	4	4	Sport	28	8	Visualization	Yoga, Campus Board
18	6	2	Bouldering	24	7	Meditation	Fingerboard, Weight Lifting
19	3	5	Sport, Trad	26	5	Mental Rehearsal	Hangboard, Core
20	9	3	Bouldering, Sport	31	6	Breathing Techniques	Fingerboard, Campus Board
21	4	3	Sport	29	5	Breathing Techniques	Fingerboard, Yoga
22	6	4	Bouldering, Sport	27	4	Visualization	Core, Hangboard
23	2	3	Sport, Trad	25	6	Visualization	Weight Lifting, Yoga
24	7	4	Bouldering, Sport	29	7	Meditation	Fingerboard, Campus Board
25	5	5	Trad, Sport	27	8	Breathing Exercises	Hangboard, Core
26	4	2	Bouldering	24	5	Positive Self-Talk	Weight Lifting, Fingerboard

Table 3. Sample Survey Responses from Female Climbers

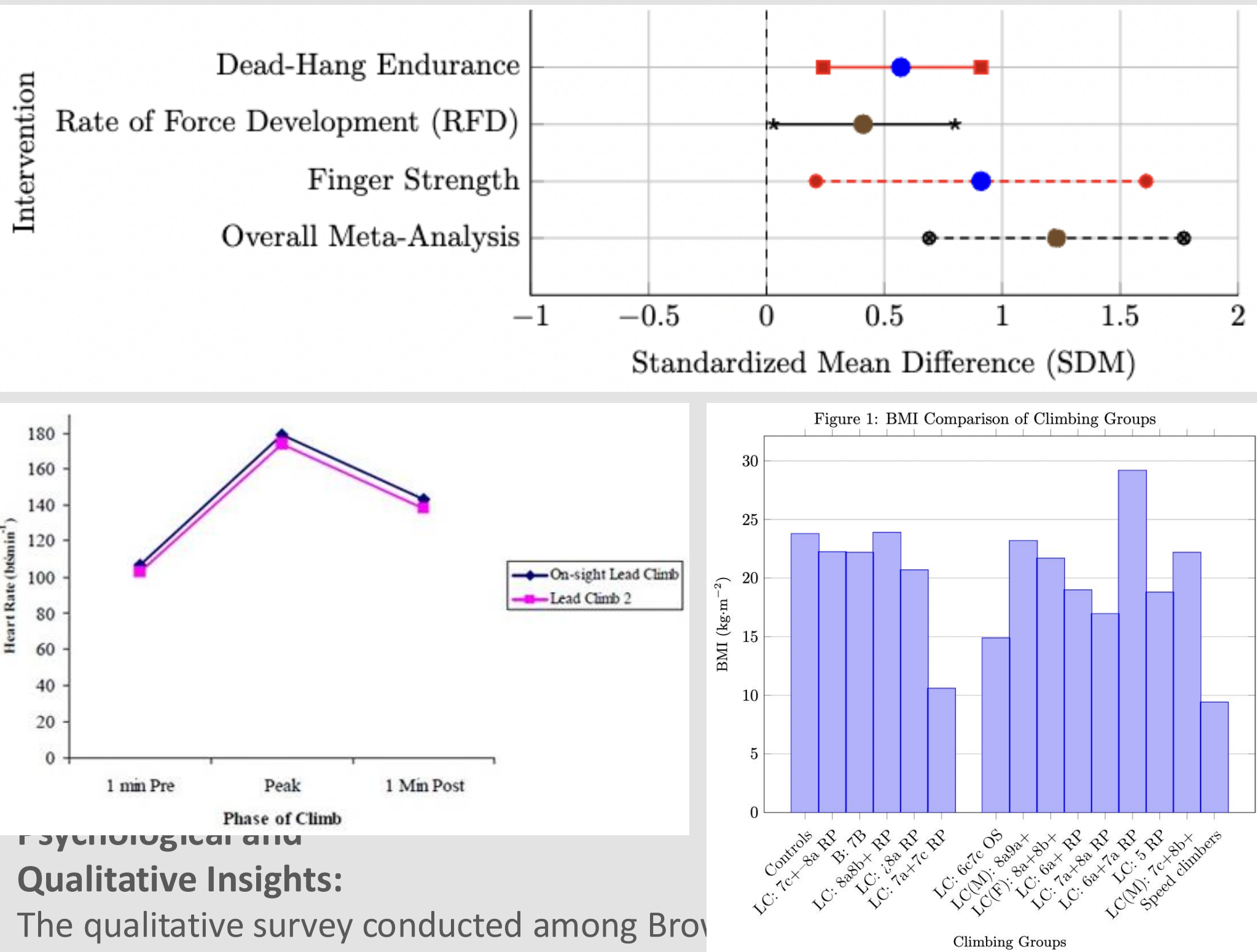
Results

Preliminary findings offer significant insights into the physiological and psychological factors influencing female rock climbers.

Quantitative Findings:

The initial quantitative analysis revealed important correlations between body composition, grip strength, and climbing performance across different skill levels. DEXA scans indicated that elite climbers (ELT) possess a higher muscle mass to body fat ratio compared to lower advanced (ADV-L) and higher advanced (ADV-H) climbers, supporting the hypothesis that a favorable strength-to-weight ratio is crucial for optimal climbing performance.

Below: suggests that climbing-specific training generally improve performance, with Dead-Hang Endurance showing the most reliable improvement.



The qualitative survey conducted among BrownGirlsClimb members provided insights into the psychological dynamics at play in climbing. A heatmap analysis of the data revealed a significant relationship between the frequency of climbing (measured in climbs per week) and reported fear of falling. Younger climbers (ages 18-25) who climbed more frequently exhibited lower levels of fear, suggesting that regular exposure to climbing environments may help mitigate anxiety. Conversely, older climbers or those with less frequent climbing sessions reported higher levels of fear, indicating a potential area for targeted psychological interventions. Survey provided narratives that highlighted strategies helpful for managing fear and anxiety. For example, the importance of mental visualization techniques and structured breathing exercises to maintain composure during challenging climbs.

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Conclusion

- Higher muscle mass to body fat ratio is positively correlated with enhanced grip strength and superior climbing outcomes, especially in elite climbers.
- For climbers, it is evident that the addition of systematic resistance-training (e.g., fingerboard, campus board, or upper-body resistance training) can yield greater improvements in climbing-specific fitness than climbing-training alone, across several performance levels.
- Grip force of elite climbers exceeds the grip force of non-climbers by far. Subsequently, careful and gradual grip force training on both hands concurrently seems to be advisable for starters in the climbing sport.
- The study acknowledges limitations, including a scarcity of research focusing solely on female climbers and variability in study methodologies.
- These limitations make generalizing findings difficult, highlighting the need for more rigorous and standardized research.
- The importance of an integrated and tailored training approach that considers both physiological and psychological dimensions may enhance performance.

Recommendations/ Future Work

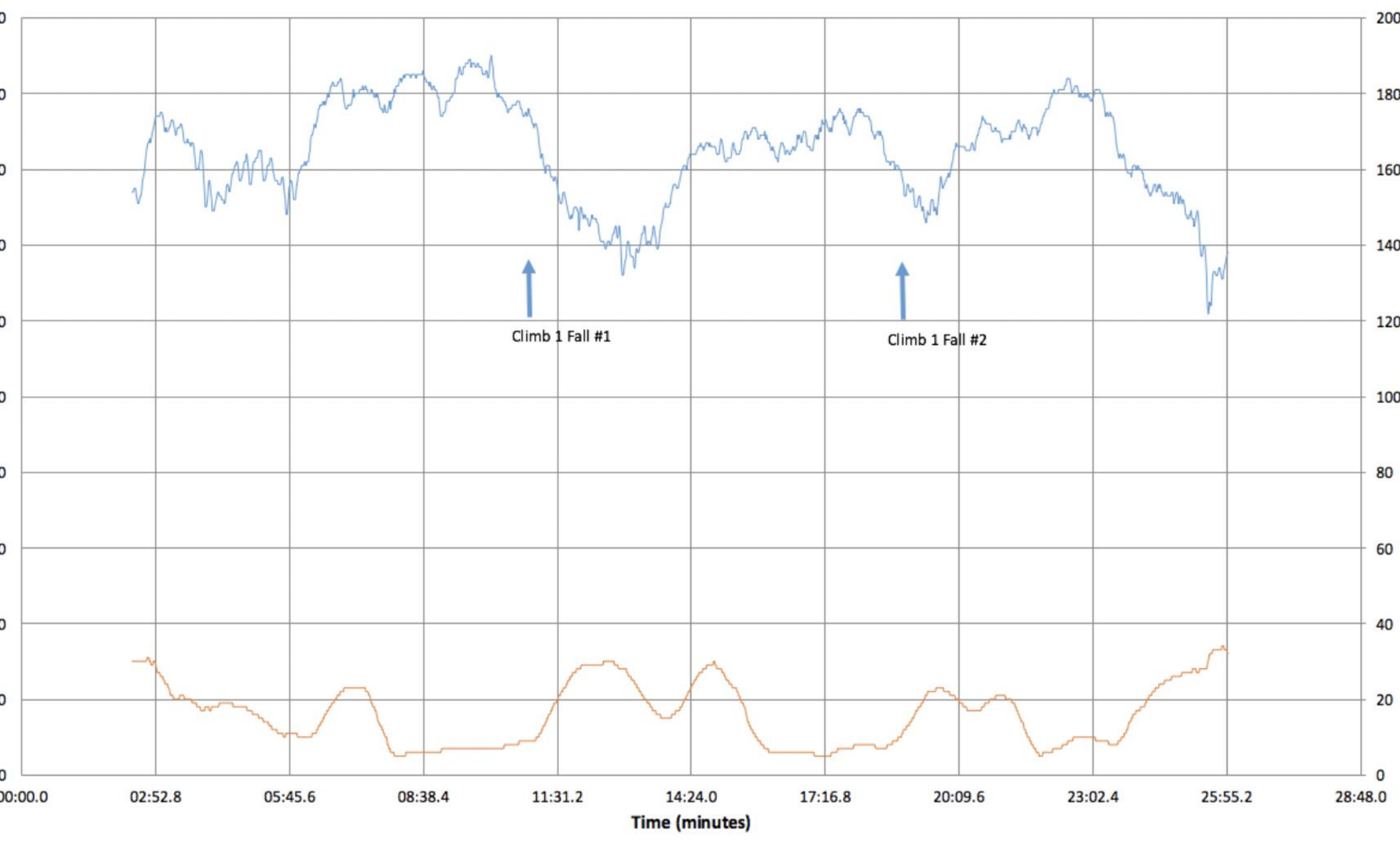
- Future research should focus on gender-specific studies for female climbers, particularly at the elite level, to better understand the unique physiological and psychological factors affecting their performance.
- Expanding the sample size and including a broader demographic of female climbers could provide more robust data and improve generalizability of results.
- Research should investigate the impact of specific training regimens, such as resistance training, combined with psychological interventions like mindfulness, on performance outcomes.
- Exploring the long-term effects of these interventions could offer valuable insights into sustainable performance enhancement strategies.
- Integrating advanced body composition analysis tools and psychological assessments into routine training could help create more effective, individualized training programs.

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Git Repo:



** HRV is a Measure of the variation in time between heartbeats